Risk-Aware Daily Allocation between Market and IPO Sleeves

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STAT 4830

What problem are we solving?

- **Motivation**: apply course concepts to the financial markets, a notoriously noisy space to optimize

Decision each day: weights for a portfolio of IPOs vs. the market

- Long-only
- Fully invested
- Weights sum to 100%
- Later expanded to sector sleeves

Why this is can be difficult:

- IPOs can have large upside
- Tail behavior is unstable
- Regimes shift quickly
- Naive return maximization over-concentrates

Loss Function

$$L = -E[r] + \lambda_{\text{cvar}} \cdot \text{CVaR}_\alpha(r) + \lambda_{\text{vol}} \cdot \sigma(r) + \lambda_{\text{div}} \cdot \text{HHI}(w) + \lambda_{\text{to}} \cdot \|\Delta w\|_1 + \lambda_{\text{bm}} \cdot \max(0,\ r_{\text{bm}} - r) + \lambda_{\text{lg}} \cdot (-\log(\text{growth}))$$

Return Maximization

maximize mean portfolio return (via minimizing negative mean return)

Tail Risk Control

smooth CVaR penalty at level \(\alpha\) (default 5%).

Volatility Penalty

Variance penalty plus extra penalty when annualized vol exceeds target.

Diversification Control

minimum-weight penalty to discourage all-in concentration

Turnover Penalty

L1 turnover penalty and L2 weight-path penalty across days

Benchmark Discipline

Penalty for underperforming benchmark return r_bm

Growth Incentive

−log(growth) term to reward portfolio growth over time

Literature Review

Online portfolio optimization: Adaptive allocation, online updates, and sensitivity to regime changes.

- Cover, 1991; Helmbold et al., 1998

IPO anomaly and post-IPO risk: High upside but fragile tails; IPOs amplify instability and timing risk.

- Ritter, 1991; Loughran & Ritter, 1995

Risk-aware objectives: CVaR, drawdown, and volatility-aware losses better match investment goals.

- Rockafellar & Uryasev, 2000

Sequence models for allocation: RNNs and attention can model temporal patterns, but objective alignment still matters.

- Lim & Zohren, 2021 and related time-series allocation work

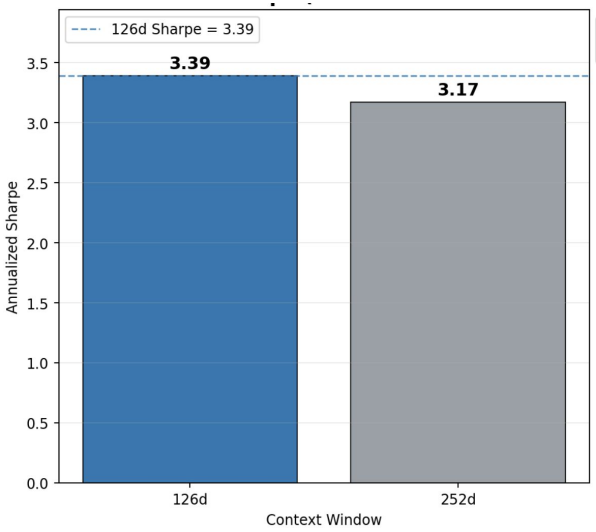
Varying Hyperparameters, batch size, and other training changes

What we varied	Examples	What we learned
Optimization settings	Learning rate, schedules, gradient clipping, checkpoint rules	Training stability mattered; clipping reduced blow-ups.
Batching choices	Batch size, sequence grouping, update cadence	Results could move materially even when architecture stayed fixed.
Regularization	Penalty weights, dropout, weight decay, turnover strength	Too little caused concentration; too much caused static allocations.
Evaluation plumbing	Early stopping, selection metric, holdout split	Selection criteria can change the “winner” without changing the model.

Varying context window, evaluation period

Context window experiments

- Short memory: Lower complexity, faster adaptation
- Long memory: More history, more noise risk
- Observed result: context length alone gave incremental gains unless paired with a better objective.
- Fit only on information available up to the day.
- Evaluate on future periods only.



The two best performing context windows

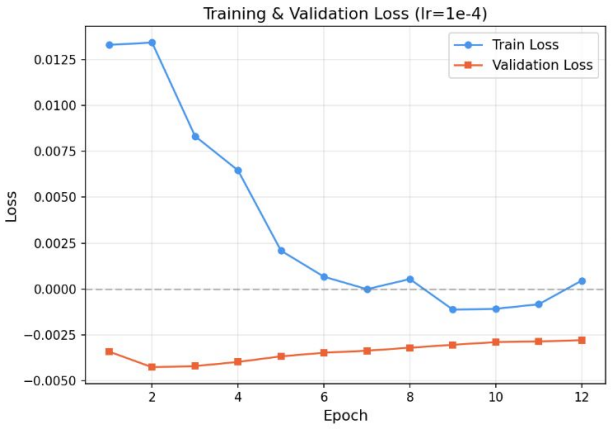
Model Architecture Testing: LSTM

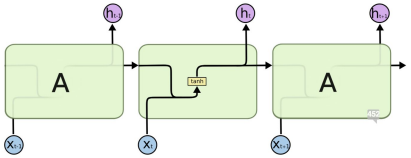
Why we tried it:

- Better long-memory handling than a simple recurrent baseline, because LSTM has a separate memory cell and extra gates

What actually happened

- Mixed gains in some branches, but not a dominant winner.
- Ultimately led to similar results as GRU

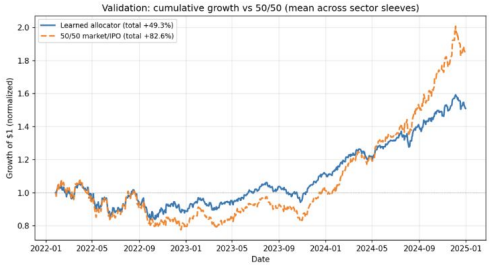




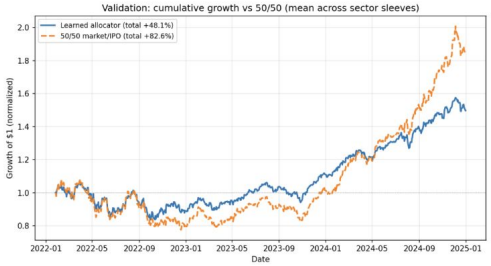
The repeating module in a standard RNN contains a single layer.

Validation: cumulative growth vs 50/50 (same scale per image)

GRU



LSTM



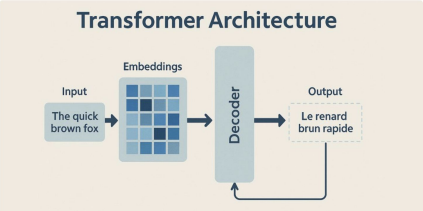
Model Architecture Testing: Transformer

Why we tried it:

- Flexible temporal interactions and richer sequence modeling.
- Hypothesis: could adapt better to changing IPO regimes.

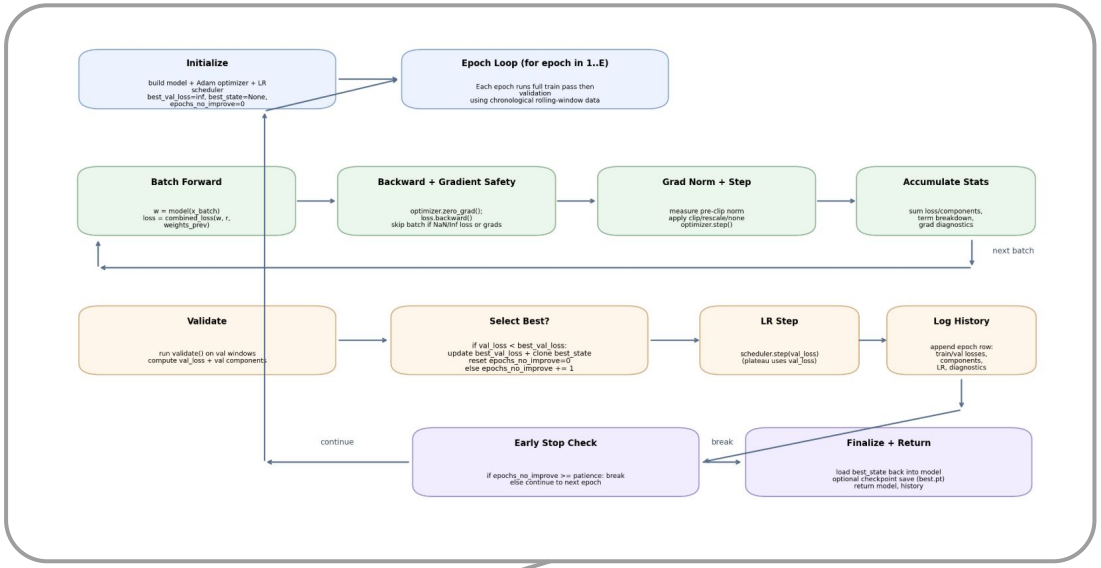
What actually happened

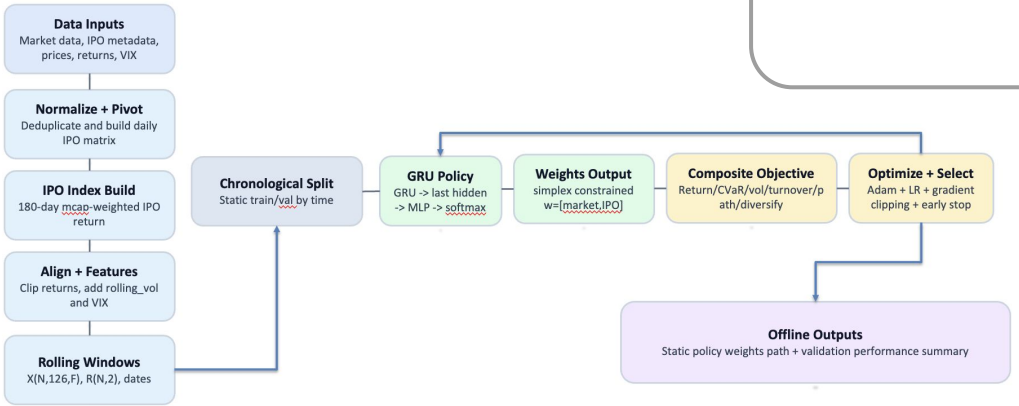
- Runtime and complexity increased materially
- Performance gains were inconsistent and ultimately underperformed the other two models
- Negative result was useful: “more expressive” was not automatically “better.”



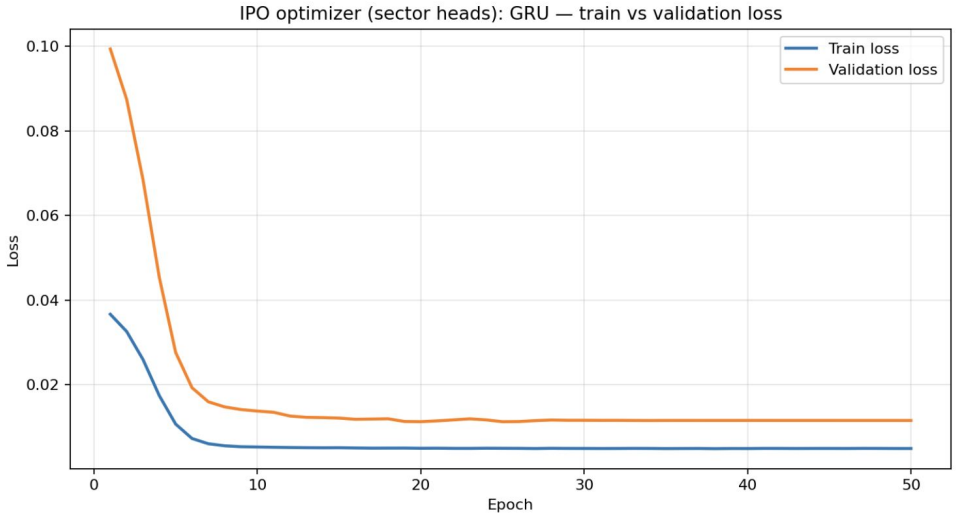
Sector	Annual Returns	Sharpe	Max Drawdown
Unknown	45.19%	1.61	-22.99%
Healthcare	49.28%	1.85	-16.46%
IT	30.37%	1.01	-37.91%
Financials	10.26%	0.56	-26.83%
Consumer	12.05%	0.49	-53.40%
Industrials	45.28%	1.45	-23.44%
Energy	7.34%	0.45	-30.71%
Communication	-21.30%	-0.38	-75.62%
Real Estate	2.41%	0.27	-54.95%
Materials	-14.35%	-0.18	-54.82%

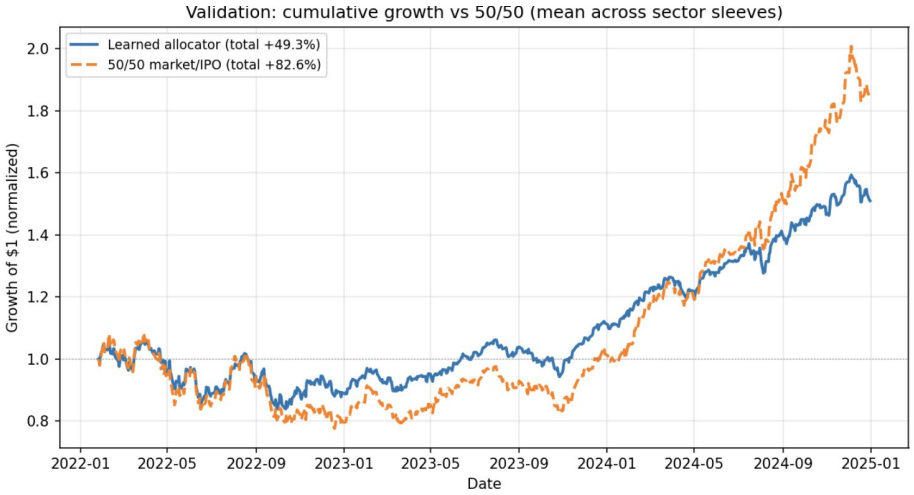
Offline GRU Policy Pipeline Architecture





Offline Model Results





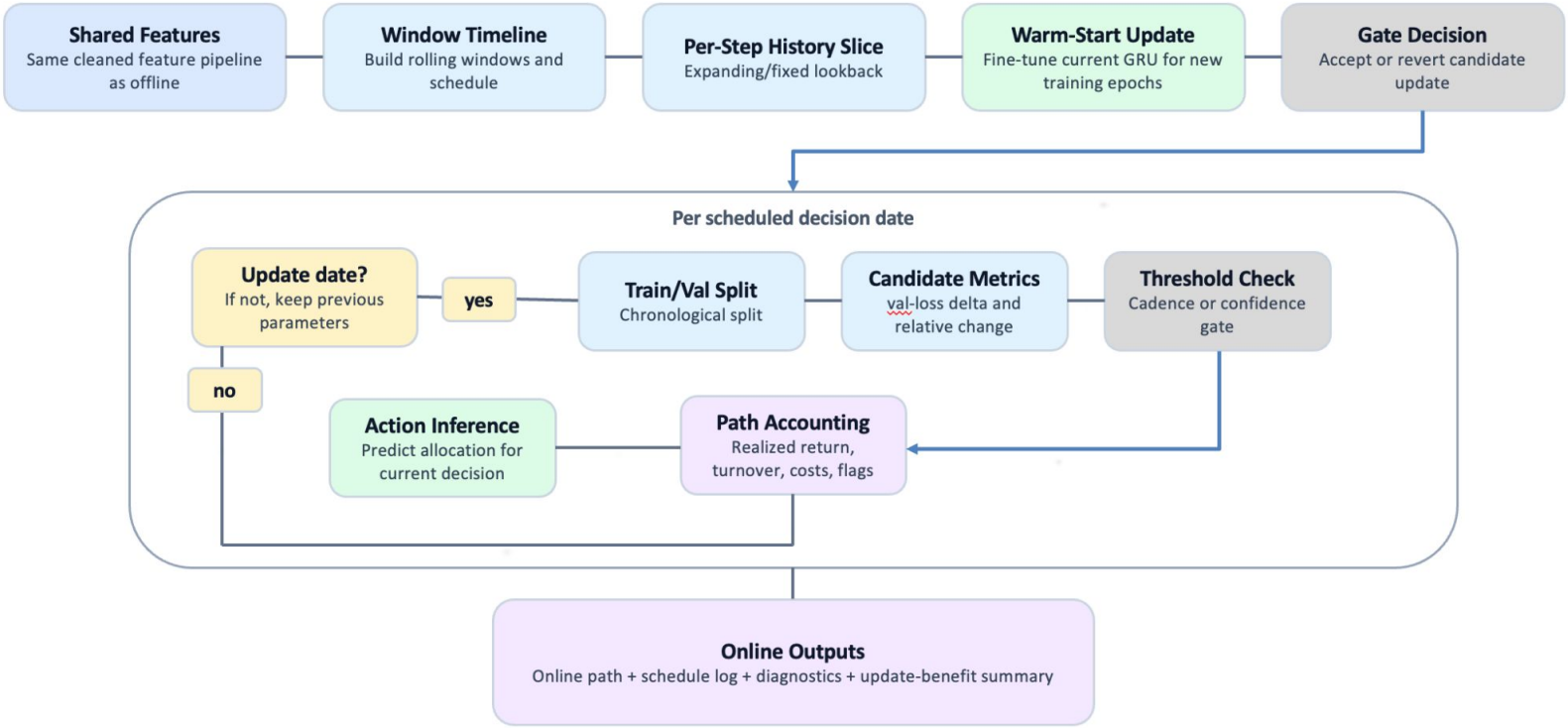
Takeaway: Training and validation loss both fall quickly and then stabilize, suggesting the GRU learns a usable allocation policy without major overfitting.

Takeaway: The learned IPO allocation grows steadily but underperforms the simple 50/50 market/IPO benchmark over the validation period.

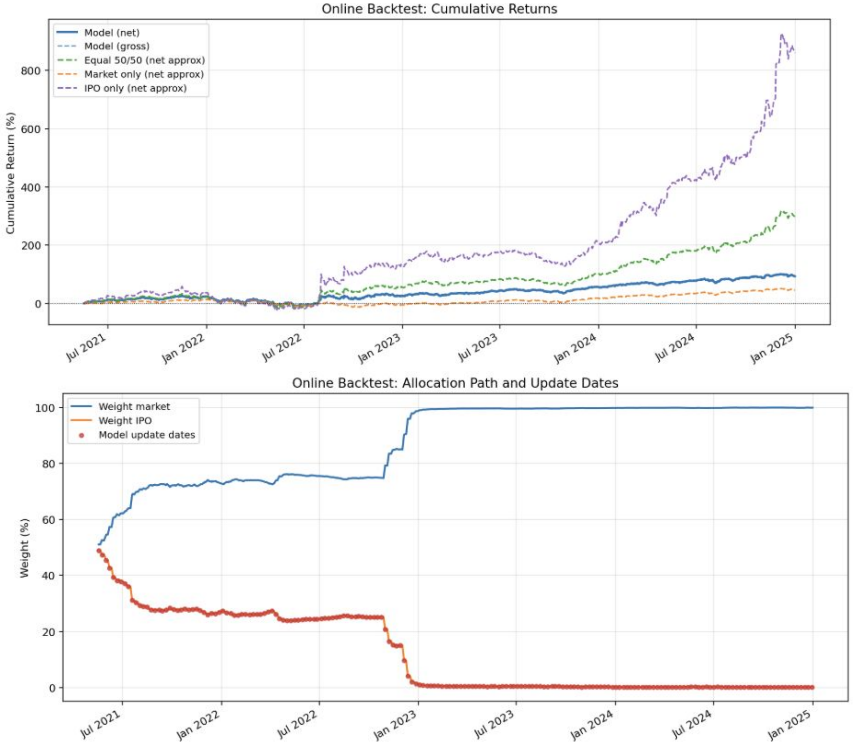
Offline Model Results

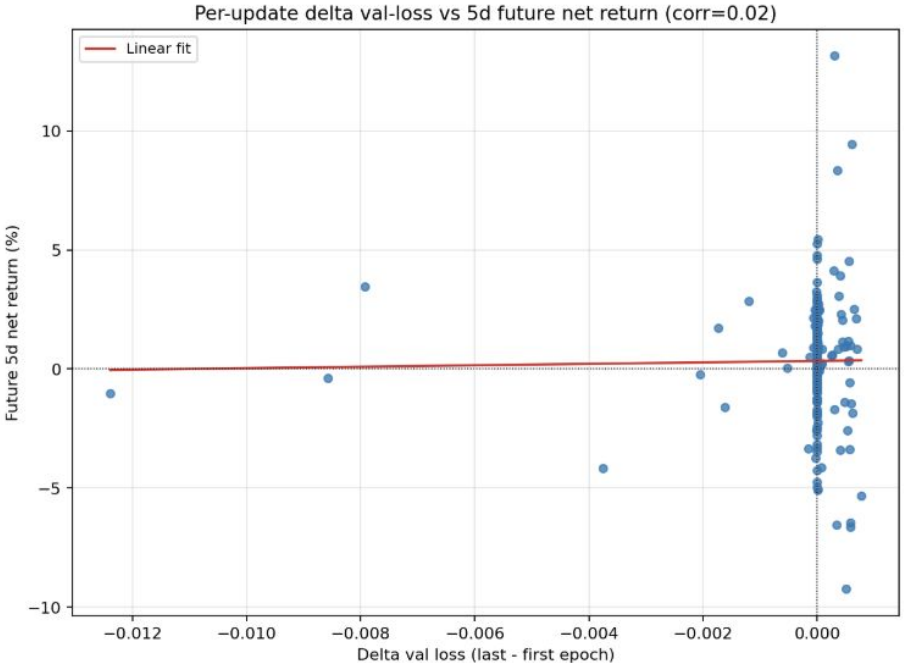
Strategy	Total return	Ann. Return	Ann. Vol	Sharpe	Max DD
Model	49.25%	45.12%	16.82%	2.30	-10.47%
Equal 50/50	82.90%	75.32%	30.86%	1.97	-25.99%
Market Only	27.07%	24.95%	14.36%	1.62	-9.97%
IPO Only	145.92%	130.89%	54.95%	1.79	-41.13%

Online GRU Policy Pipeline Architecture

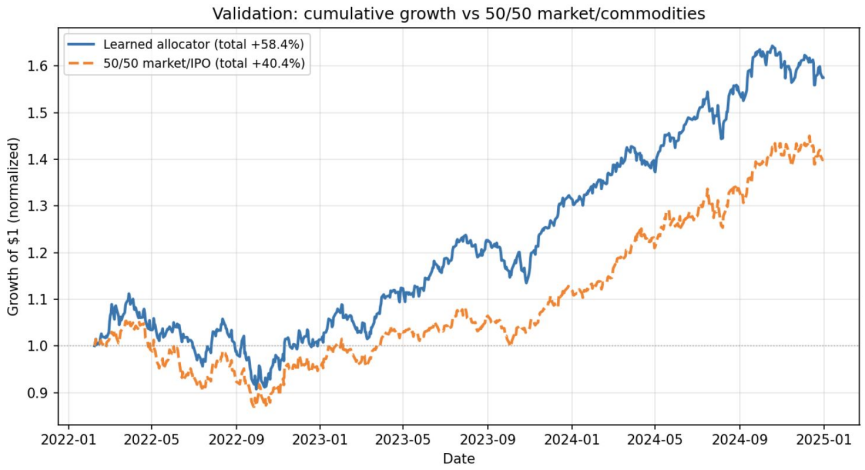


Online Policy Optimization



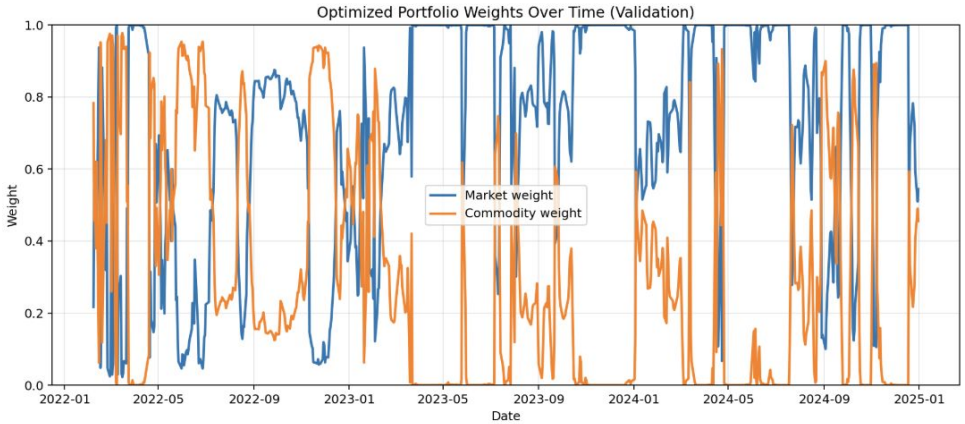


Commodities GRU Model Results



The learned commodities allocator outperforms the static 50/50 benchmark, indicating the GRU adds value through dynamic allocation.

The model shifts aggressively between market and commodities, showing that it is actively timing exposures rather than holding a fixed mix.



Takeaways & Next Steps

Worked:

- Risk/diversification shaping reduced concentration collapse.
- Sector and VIX data
- Gradient clipping improved optimization stability.

Mixed / limited:

- Changing our objective function to account for longer term wealth accumulation
- Context-window tuning alone gave mostly incremental gains.
- Sector-aware modeling helped diagnosis but raised metadata dependence.
- Next steps: Implement cadence/threshold check to prevent training on noise

Failed:

- Transformer complexity was not justified in this regime.
- Some settings improved loss but worsened realized path quality.
- Over-regularization sometimes killed useful adaptation.
- Integration of shorting capacity in policy

Thank you for listening!